

Line-preserving deformations of Fano flows: a universal span identity and a Petersen obstruction

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AI-assistance disclosure. OpenAI Codex agents (GPT-5-series models, accessed in July 2026) developed the fixed-line formulation, found the telescoping potential, derived the two-cycle normal form and the Petersen example, wrote the search and checking programs, and drafted this manuscript under human direction. The checkers and the present proof exposition are themselves AI-assisted; “independent” below means logically separate code, not independent human verification. No human author or referee is represented as having certified the results. This draft must not be submitted until a human author has checked every proof and computation, verified the literature search, and accepted ordinary scholarly responsibility. The results are partial structural statements and do not prove or disprove the five-cycle double cover conjecture.

Abstract

Let f be a nowhere-zero $\mathbb{F}_2^3$ -flow on a finite loopless cubic graph and fix a nonzero functional μ . Its kernel determines a Fano line L , a spanning factor F_L , and a binary vector recording the components of F_L whose four affine boundary colours all occur oddly. For each of the three nonzero values $t \in L$, binary cycles avoiding the value class $f^{-1}(t)$ give an initially valid line-preserving switch map. We prove that the sum of the images of these three linear maps always contains the defect vector. The proof is a short cut-cycle duality argument closed by an explicit quadratic potential which telescopes against flow conservation.

The span identity is only a first-order statement. We give an exact nonlinear normal form: flows with the same μ -projection are parametrised by two binary cycles p, q , and line cleaning is equivalent to their covering the kernel factor together with quadratic product-parity equations on its components. This yields a general obstruction when the kernel factor is a perfect matching whose simple contraction Q satisfies $|E(Q)| < 4 \operatorname{girth}(Q)$.

Finally, we display a nowhere-zero $\mathbb{F}_2^3$ -flow on the Petersen graph with two prescribed lines that cannot be cleaned while their functional projections are fixed. The obstruction has a finite human-checkable proof: in both cases the matching contraction is K_5 , and a clean lift would force a Tait colouring of the Petersen graph. The same graph has an explicit five-cycle double cover, so the example is not a counterexample to that conjecture. The manuscript and all calculations are AI-assisted and await independent human review.

1 Scope and background

All graphs in the main results are finite, undirected, loopless, and cubic; parallel edges are allowed unless simplicity is stated. Since the coefficient groups have exponent two, orientations play no role. An $\mathbb{F}_2^k$ -flow is a map $f : E(G) \rightarrow \mathbb{F}_2^k$ satisfying

$$\bigoplus_{e \ni v} f(e) = 0 \quad (v \in V(G)).$$

It is *nowhere-zero* if $f(e) \neq 0$ on every edge. At a cubic vertex the three values of a nowhere-zero $\mathbb{F}_2^3$ -flow are the three points of a line of the Fano plane.

Fano colourings of cubic graphs and their relation to bridgelessness, perfect matchings, and the Fulkerson and Fan–Raspauld conjectures are well established [2, 7, 6, 4, 5]. In particular, the existence of nowhere-zero $\mathbb{F}_2^3$ -flows on bridgeless graphs follows from Jaeger’s 8-flow theorem [3]. This paper does not claim any of that background as new.

Our question is narrower. We fix one nonzero functional $\mu : \mathbb{F}_2^3 \rightarrow \mathbb{F}_2$, preserve the binary flow $\mu \circ f$, and ask whether the corresponding kernel line can be made *clean*, in the sense defined below. This is not the established k -line Fano-flow problem, which counts the vertex lines used by a flow. It is also different from recent work on reconfiguring arbitrary nowhere-zero flows by cycle-supported steps [1].

There is a relevant recent neighbour. Mkrtchyan studies non-conflicting $\mathbb{F}_2^2$ -flows on contractions of complementary 2-factors and proves that the Petersen graph is an obstruction [8, Proposition 2.21]. Our clean-lift condition, fixed functional projection, combined-image identity, and quadratic normal form are different. Thus the general fact that “Petersen is an obstruction to a constrained flow extension” is not new; only the specific statements proved here are provisionally claimed as new.

For further separation from the motivating conjecture, a five-cycle double cover (5-CDC) is a list of five even edge-subsets in which every edge appears exactly twice. Král’ et al. give the equivalent Desargues-configuration colouring formulation [6, Theorem 7.1]. Nothing below proves that every bridgeless graph has such a cover. Section 6 gives a 5-CDC of our Petersen example explicitly.

2 A fixed line and its switch images

Let $f : E(G) \rightarrow \mathbb{F}_2^3 - \{0\}$ be a flow and fix a nonzero functional μ . Write

$$K = \ker \mu, \quad L = K - \{0\}, \quad A = \{a \in \mathbb{F}_2^3 : \mu(a) = 1\}.$$

The three-element set L is a Fano line and A is its affine complement. Put

$$F_L = f^{-1}(L).$$

At each cubic vertex, an even number of incident values lie in A . Consequently every vertex has degree one or three in the spanning subgraph F_L . Let $\mathcal{K}$ be its set of connected components.

For a component $W \in \mathcal{K}$, flow conservation across its shore gives

$$\bigoplus_{e \in \delta(W)} f(e) = 0.$$

The boundary edges all have values in A . The parities of the four affine value multiplicities on $\delta(W)$ are equal. One quick proof is to write $A = b + K$: the kernel of the 3×4 matrix whose columns are $(1, k)$, $k \in K$, is generated by the all-one vector. Define the *rainbow-defect vector*

$$r \in \mathbb{F}_2^{\mathcal{K}}, \quad r_W = 1 \iff \text{each of the four values in } A \text{ occurs oddly on } \delta(W).$$

We call L *clean* if $r = 0$.

For $t \in L$, translation by t partitions A into two two-element orbits. Choose one orbit P_t , and put

$$M_t = f^{-1}(t), \quad Z_t = Z_1(G - M_t; \mathbb{F}_2).$$

Thus Z_t is the binary cycle space of the graph obtained by deleting the t -valued edges. For $C \in Z_t$, set

$$\tau_t(C)_W = |C \cap \delta(W) \cap f^{-1}(P_t)| \pmod{2}. \quad (2.1)$$

The complementary orbit gives the same bit, because a binary cycle crosses every cut evenly.

Adding t to the values on C preserves flow conservation. Because $C \cap M_t = \emptyset$, it creates no zero edge. Formula (2.1) is the exact defect update for this single switch, computed at the displayed initial flow. Define the combined image

$$\mathcal{W}_L = \sum_{t \in L} \text{im } \tau_t. \quad (2.2)$$

3 The combined-image theorem

Theorem 3.1 (combined-line span). *For every finite loopless cubic G , every nowhere-zero $\mathbb{F}_2^3$ -flow f , and every nonzero functional μ ,*

$$r \in \mathcal{W}_L. \quad (3.1)$$

Proof. Suppose that (3.1) fails. By finite-dimensional linear duality there is $y \in \mathbb{F}_2^K$ such that

$$y \cdot r = 1, \quad y \cdot \tau_t(C) = 0 \quad (t \in L, C \in Z_t). \quad (3.2)$$

Let $\mathcal{Y} = \text{supp } y$ and

$$U = \bigcup_{W \in \mathcal{Y}} V(W).$$

For each t , equation (2.1) gives

$$y \cdot \tau_t(C) = |C \cap \delta(U) \cap f^{-1}(P_t)| \pmod{2}. \quad (3.3)$$

In any graph, the cut space is the orthogonal complement of the cycle space. Hence

$$R_t = \delta(U) \cap f^{-1}(P_t) \quad (3.4)$$

is a cut of $G - M_t$. Choose a vertex indicator $x_t : V(G) \rightarrow \mathbb{F}_2$ for this cut.

Apply an invertible change of coordinates in $\mathbb{F}_2^3$ and label vectors by binary integers with

$$1 = (1, 0, 0), \quad 2 = (0, 1, 0), \quad 4 = (0, 0, 1).$$

We may assume

$$L = \{1, 2, 3\}, \quad A = \{4, 5, 6, 7\}, \quad P_t = \{4, 4 + t\}. \quad (3.5)$$

Let u be the indicator of U , and at each vertex put

$$q = (u, x_1, x_2, x_3). \quad (3.6)$$

Across an edge, (3.4)–(3.5) allow the zero difference and the following single nonzero difference:

$f(e)$	$q_v + q_w$
1	(0; 100)
2	(0; 010)
3	(0; 001)
4	(1; 111)
5	(1; 100)
6	(1; 010)
7	(1; 001)

(3.7)

For example, a line-valued edge has both ends in the same component of F_L , so its u -difference is zero. If its value is t , only the cut x_t is unconstrained because that edge was deleted from $G - M_t$. On an affine edge, (3.4) says that the x_t -difference is the u -difference precisely when its colour lies in P_t . This proves every row of (3.7).

Define the potential

$$\mathcal{A}(u, x_1, x_2, x_3) = (u(x_2 + x_3), u(x_1 + x_3), x_1x_2 + x_1x_3 + x_2x_3). \quad (3.8)$$

All operations are in $\mathbb{F}_2$. Direct substitution in the seven rows of (3.7) gives

$$(\mathcal{A}(q_v) + \mathcal{A}(q_w)) \cdot f(vw) = \begin{cases} 1, & f(vw) = 4 \text{ and } u(v) + u(w) = 1, \\ 0, & \text{otherwise.} \end{cases} \quad (3.9)$$

For transparency, the seven nonzero cases reduce as follows:

$f(e)$	$q_v + q_w$	$(\mathcal{A}(q_v) + \mathcal{A}(q_w)) \cdot f(e)$
1	(0; 100)	0
2	(0; 010)	0
3	(0; 001)	0
4	(1; 111)	1
5	(1; 100)	0
6	(1; 010)	0
7	(1; 001)	0

The zero-difference cases are immediate. As one sample expansion, in the colour-4 row all three x_i and u toggle; the change of $x_1x_2 + x_1x_3 + x_2x_3$ is 1, and $4 = (0, 0, 1)$. In the colour-5 row the changes of the first and third coordinates of $\mathcal{A}$ are both $x_2 + x_3$, and $5 = (1, 0, 1)$, so they cancel. The remaining rows follow by the same two-line expansion.

Sum (3.9) over all edges and regroup the left side at vertices:

$$\begin{aligned} \sum_{vw \in E(G)} (\mathcal{A}(q_v) + \mathcal{A}(q_w)) \cdot f(vw) &= \sum_{v \in V(G)} \mathcal{A}(q_v) \cdot \left(\bigoplus_{e \ni v} f(e) \right) \\ &= 0 \end{aligned} \quad (3.10)$$

by flow conservation. The right side of (3.9) sums to

$$|\delta(U) \cap f^{-1}(4)| \pmod{2}. \quad (3.11)$$

On the other hand, summing the colour-4 boundary parity over the components selected by y cancels affine edges joining two selected components and leaves exactly (3.11). By the definition of r , this is $y \cdot r = 1$, contradicting (3.10). $\square$

Corollary 3.2 (at most two components). *If F_L has at most two components, then L is clean or one valid line-preserving binary-cycle switch makes it clean.*

Proof. Every vector in every $\text{im } \tau_t$ has even Hamming weight: after summing (2.1) over all components, every counted affine edge is counted at both ends. If F_L has one component, its even-weight subspace is zero, so Theorem 3.1 gives $r = 0$. If it has two components, that subspace is one-dimensional. When $r \neq 0$, some $\text{im } \tau_t$ in (2.2) must be nonzero and therefore contains r . One cycle producing r cleans the line. A single cycle has no overlap correction and avoids M_t , so the switch remains nowhere-zero. $\square$

4 The exact quadratic normal form

Theorem 3.1 combines three maps computed at the initial flow. It does not assert that the corresponding three switches can be performed simultaneously. An affine edge used by two switches has a quadratic correction, and two switches may also turn a line-valued edge into zero. We now describe the exact problem without switch-order ambiguity.

Let

$$h = \mu \circ f, \quad F_\mu = h^{-1}(0).$$

Thus $F_\mu = F_L$. Choose $b \in \mathbb{F}_2^3$ with $\mu(b) = 1$ and identify $K = \ker \mu$ with $\mathbb{F}_2^2$.

Proposition 4.1 (two-cycle normal form). *The flows f' satisfying $\mu \circ f' = h$ are exactly*

$$f'_e = b h_e + (p_e, q_e), \quad (4.1)$$

where $p, q \in Z_1(G; \mathbb{F}_2)$. Such a flow is nowhere-zero exactly when

$$p_e \vee q_e = 1 \quad (e \in F_\mu). \quad (4.2)$$

It is clean on every component W of F_μ exactly when

$$\sum_{e \in \delta(W)} p_e q_e = 0 \quad (W \in \mathcal{K}). \quad (4.3)$$

Proof. If $\mu \circ f' = h$, then

$$s = f' + b h$$

is a K -valued flow. Its two coordinates are binary cycles p, q , which proves existence and uniqueness in (4.1). Conversely, any two binary cycles in (4.1) give a flow with projection h .

On an edge with $h_e = 1$, the value $b + (p_e, q_e)$ is automatically nonzero because it lies outside K . On an edge of F_μ , it is (p_e, q_e) , proving (4.2).

The boundary of a component W consists of affine edges. Its size is even, and each binary cycle has even intersection with its cut. The parity of the affine colour b , whose residual coordinate is $(0, 0)$, is

$$\begin{aligned} \sum_{e \in \delta(W)} (1 + p_e)(1 + q_e) &= |\delta(W)| + \sum_{\delta(W)} p_e + \sum_{\delta(W)} q_e + \sum_{\delta(W)} p_e q_e \\ &= \sum_{e \in \delta(W)} p_e q_e. \end{aligned} \quad (4.4)$$

As in Section 2, flow conservation makes the four affine colour parities equal. Thus they are all even exactly when (4.3) holds. $\square$

Remark 4.2. Equations (4.2)–(4.3) say that p, q cover the kernel factor and that, after its components are contracted, the affine edge set $\{e : p_e = q_e = 1\}$ has even degree at every contracted vertex. The cover clauses are monotone, but the boundary clauses contain the quadratic products $p_e q_e$. Theorem 3.1 removes the corresponding first-order obstruction; it does not solve these quadratic equations.

Corollary 4.3 (Tait graphs have no fixed-projection obstruction). *Let G be a Tait-colourable cubic graph. For every binary cycle $h \in Z_1(G; \mathbb{F}_2)$, the two-cycle conditions (4.2)–(4.3) are feasible for $F_\mu = h^{-1}(0)$. Consequently every functional projection has a clean nowhere-zero $\mathbb{F}_2^3$ -lift.*

Proof. Fix a nowhere-zero $\mathbb{F}_2^2$ -flow $s = (p, q)$ supplied by a Tait colouring. Since s is nowhere-zero on every edge, p, q cover F_μ , proving (4.2).

Let W be a component of F_μ . Every edge of $\delta(W)$ has $h_e = 1$, and the binary cycle h crosses the cut evenly, so $|\delta(W)|$ is even. Let a, b, c be the parities on this cut of the three nonzero s -values $(1, 0), (0, 1), (1, 1)$, respectively. Flow conservation across the cut gives

$$a + c = 0, \quad b + c = 0,$$

so $a = b = c$. But

$$0 = |\delta(W)| = a + b + c = c \quad \text{in } \mathbb{F}_2.$$

Thus the $(1, 1)$ -colour occurs evenly on $\delta(W)$, which is exactly $\sum_{e \in \delta(W)} p_e q_e = 0$. This proves (4.3), and Proposition 4.1 supplies the clean lift. $\square$

5 A structural matching-contraction obstruction

Recall that a cubic graph is *Tait-colourable* if it has a proper three-edge-colouring. Equivalently, it admits a nowhere-zero $\mathbb{F}_2^2$ -flow.

Theorem 5.1 (simple-contraction obstruction). *Suppose F_μ is a perfect matching of a cubic graph G , and let $Q = G/F_\mu$. If*

$$Q \text{ is simple,} \quad |E(Q)| < 4 \text{ girth}(Q), \quad G \text{ is not Tait-colourable,} \quad (5.1)$$

then no clean nowhere-zero flow f' can have $\mu \circ f' = \mu \circ f$.

Proof. Assume that a clean lift f' exists. The edges outside F_μ become the edges of Q . Their four affine colours partition $E(Q)$. Cleanliness says that every colour class has even degree at each vertex of Q , so each is an even subgraph.

Every nonempty even subgraph of a finite simple graph contains a cycle, and therefore has at least $\text{girth}(Q)$ edges. The strict inequality in (5.1) implies that one of the four affine colours is absent. Choose this absent colour as the base b in (4.1), and write

$$f' = b h + s, \quad s : E(G) \rightarrow K.$$

On an affine edge, $s \neq 0$, since $f' \neq b$. On an edge of F_μ , $h = 0$ and $s = f' \neq 0$. Hence s is a nowhere-zero $K \cong \mathbb{F}_2^2$ -flow on all of G , which is a Tait colouring. This contradicts (5.1). $\square$

6 The Petersen fixed-line obstruction

We now give all data needed for a finite example. The graph has vertices $0, \dots, 9$; its edges, edge identifiers, and flow values are:

id	edge	$f(e)$	id	edge	$f(e)$
0	0–2	6	8	3–5	5
1	0–4	3	9	3–8	7
2	0–6	5	10	4–5	6
3	1–4	5	11	5–9	3
4	1–7	1	12	6–8	3
5	1–8	4	13	6–9	6
6	2–3	2	14	7–9	5
7	2–7	4			

(6.1)

The three incident values at vertices $0, \dots, 9$ are, respectively,

$$\begin{aligned} (6, 3, 5), (5, 1, 4), (6, 2, 4), (2, 5, 7), (3, 5, 6), \\ (5, 6, 3), (5, 3, 6), (1, 4, 5), (4, 7, 3), (3, 6, 5). \end{aligned} \tag{6.2}$$

Each triple has XOR zero, so (6.1) is a nowhere-zero $\mathbb{F}_2^3$ -flow. The edge list is a labelling of the Petersen graph; its nauty-canonical graph6 encoding is

IsP00kWHG.

Theorem 6.1 (two failed prescribed lines). *For the flow (6.1), neither $L = \{1, 2, 3\}$ nor $L = \{1, 6, 7\}$ can be made clean by a nowhere-zero $\mathbb{F}_2^3$ -flow having the same corresponding functional projection.*

Proof. For $L = \{1, 2, 3\}$, the kernel factor consists of edge identifiers

$$\{1, 4, 6, 11, 12\}; \tag{6.3}$$

for $L = \{1, 6, 7\}$, it consists of

$$\{0, 4, 9, 10, 13\}. \tag{6.4}$$

In each case the five edges are pairwise disjoint and cover all ten vertices. Contracting them in the graph (6.1), the remaining ten edges join every pair of contracted vertices exactly once. Thus $Q = K_5$.

The Petersen graph is not Tait-colourable. For a self-contained finite check, its perfect matchings in the edge indexing (6.1) are exactly

$$\begin{aligned} (0, 3, 8, 12, 14), \quad (0, 4, 9, 10, 13), \\ (1, 4, 6, 11, 12), \quad (1, 5, 7, 8, 13), \\ (2, 3, 7, 9, 11), \quad (2, 5, 6, 10, 14). \end{aligned} \tag{6.5}$$

Exhaustiveness follows by branching on the three edges incident with vertex 0; each initial choice has the two completions displayed in (6.5). Direct deletion shows that the complement of every matching in (6.5) is two 5-cycles. In a Tait colouring, one colour class is a perfect matching and its complement alternates the other two colours on even cycles, a contradiction.

Now K_5 is simple, has ten edges, and has girth three, so

$$10 < 4 \cdot 3.$$

Theorem 5.1 applied to (6.3) and (6.4) proves both claims. □

Remark 6.2 (not a 5-CDC counterexample). The following five Eulerian edge-id sets cover every edge of (6.1) exactly twice:

$$\begin{aligned} \{0, 1, 3, 4, 7\}, \\ \{3, 5, 8, 9, 10\}, \\ \{0, 2, 6, 8, 11, 13\}, \\ \{1, 2, 4, 5, 10, 11, 12, 14\}, \\ \{6, 7, 9, 12, 13, 14\}. \end{aligned} \tag{6.6}$$

Even degree at every vertex and double coverage can both be checked directly from (6.1). Thus (6.6) is a positive 5-CDC of the example.

Remark 6.3 (the surviving existential branch). The other five lines of the displayed flow have clean lifts with fixed functional projection. A complete classification of the 64 possible binary projections on the Petersen graph finds 57 feasible projections; the failures are the zero cycle and the six ten-edge complements of the perfect matchings in (6.5). The six nonzero failures have rank five and their only dependence is their total XOR. Consequently, the three-dimensional projection space of any nowhere-zero $\mathbb{F}_2^3$ -flow on the Petersen graph contains at most three of them, so at least four of its seven lines are cleanable. This finite classification is supplied as a reproducible computational audit, not needed for Theorem 6.1.

7 Reproducibility and logical status

The companion directory

```
search/fano-two-cycle-petersen-countermodel-20260726/
```

contains the canonical graph, the construction table, a semantic checker, two ordinary CNF instances, and DRAT proofs. From the repository root, run

```
python3 search/fano-two-cycle-petersen-countermodel-20260726/\
independent_checker.py
python3 search/fano-two-cycle-petersen-countermodel-20260726/\
verify_package.py
```

The first program independently enumerates the 64 binary cycles and all 4096 ordered pairs (p, q) for each line. For each failed line, 960 pairs cover the five matching edges and none satisfies all five product-parity equations. The second program reconstructs two 65-variable, 210-clause CNFs and checks their DRAT certificates when `drat-trim` is available.

These computations are redundant for the mathematical nonexistence proof: Theorem 6.1 follows from the displayed flow, the two matching contractions, the elementary edge-count in K_5 , and the six-matchings proof that Petersen is non-Tait. The computational package serves to catch transcription errors and to support the broader classification in the last remark.

8 What is and is not new

The literature search for this draft covered the primary sources cited above and keyword/full-text searches through 26 July 2026. We did not find the combined-image theorem, the potential (3.8), the exact fixed-projection normal form (4.1)–(4.3), or the simple-contraction obstruction in those sources. We therefore regard these as *plausibly new*, not as established priority claims.

The broader setting is not new: Fano colourings, line counts, their matching interpretations, and constrained $\mathbb{F}_2^2$ -flows on contractions all have substantial prior literature. Most importantly, Mkrtchyan’s distinct Petersen obstruction means that the appearance of the Petersen graph itself carries little novelty. The potentially publishable unit is instead the following coherent package:

1. a universal first-order span identity with an explicit telescoping certificate;
2. an exact quadratic formulation explaining why that identity does not automatically integrate to a valid switch sequence; and
3. a structural contraction obstruction, instantiated by a small fully checkable fixed-projection counterexample.

This is suitable at most as a short technical note until expert review confirms the literature boundary and a human author independently verifies the proofs. It is not a resolution, a claimed breakthrough on 5-CDC, or evidence against 5-CDC: the example positively satisfies the conjectured conclusion.

Acknowledgement and responsible authorship note

This draft was produced in an autonomous-conjecture-resolution project directed by the user. AI assistance was substantive at every stage, including conjecture formation, proof discovery, computation, literature triage, and writing. Before any public submission, a human author should rederive the arguments, check the primary literature directly, rerun the artifacts in a clean environment, decide whether the contribution merits dissemination, and take responsibility for the final text. The model provider and model are tools, not authors.

References

- [1] Daniel W. Cranston, Jiaao Li, Bo Su, Zhouningxin Wang, and Ningyan Xu. Reconfiguration of nowhere-zero flows, 2026.
- [2] Fred C. Holroyd and Martin Škoviera. Colouring of cubic graphs by Steiner triple systems. *Journal of Combinatorial Theory, Series B*, 91(1):57–66, 2004.
- [3] François Jaeger. Flows and generalized coloring theorems in graphs. *Journal of Combinatorial Theory, Series B*, 26(2):205–216, 1979.
- [4] Ligang Jin, Giuseppe Mazzuoccolo, and Eckhard Steffen. Cores, joins and the Fano-flow conjectures. *Discussiones Mathematicae Graph Theory*, 38:165–175, 2018.
- [5] Ján Karabáš, Edita Máčajová, Roman Nedela, and Martin Škoviera. Cubic graphs with colouring defect 3. *The Electronic Journal of Combinatorics*, 31(2):P2.6, 2024.
- [6] Daniel Král, Edita Máčajová, Ondřej Pangrác, André Raspaud, Jean-Sébastien Sereni, and Martin Škoviera. Projective, affine, and abelian colorings of cubic graphs. *European Journal of Combinatorics*, 30(1):53–69, 2009.
- [7] Edita Máčajová and Martin Škoviera. Fano colourings of cubic graphs and the Fulkerson conjecture. *Theoretical Computer Science*, 349(1):112–120, 2005.
- [8] Vahan Mkrtchyan. Non-conflicting nowhere-zero $\mathbb{Z}_2 \times \mathbb{Z}_2$ -flows in cubic graphs. *Australasian Journal of Combinatorics*, 91(3):392–413, 2025.